NEET UG CHEMISTRY COMPLETE GUIDE | CLASS 12 PHYSICAL CHEMISTRY | HIGH-YIELD EXAM FOCUS

Chemical Kinetics

A Comprehensive, Exam-Oriented Master Tutorial with NEET Practice Assessments & Complete Step-by-Step Solutions

1. Syllabus Overview & Learning Objectives

Chemical Kinetics is one of the highest-yielding topics in Physical Chemistry for NEET UG, contributing 2 to 3 questions each year. While chemical thermodynamics establishes whether a transformation is thermodynamically feasible (spontaneous when $\Delta G < 0$) and chemical equilibrium predicts the extent to which it proceeds, chemical kinetics governs the **speed (reaction velocity)** of the chemical change, the pathway (mechanism), and the molecular parameters controlling it.

NEET LEARNING CHECKLIST

- Differentiate between average and instantaneous rates of chemical reactions using stoichiometric expressions.
- Formulate differential rate equations and deduce the overall order and individual reactant orders from experimental rate laws.
- Distinguish sharply between **order** and **molecularity** of elementary and complex reactions.
- Identify the slowest step as the **rate determining step (RDS)** in multi-step reaction mechanisms.
- Derive and apply integrated rate equations and half-life formulas for **zero-order** and **first-order** processes.
- Master gaseous first-order reactions involving total system pressure changes.
- Understand pseudo-first-order reactions and why abundant solvents lead to apparent first-order kinetics.
- Analyze the temperature dependence of rate constants using the **Arrhenius equation**, Maxwell-Boltzmann energy distribution curves, and activation energy (E_a).
- Understand how a catalyst creates an alternative lower activation energy pathway without altering ΔG or equilibrium constant K_{eq}.
- Explain the foundational concepts of **Collision Theory**: collision frequency (Z), threshold energy, steric/probability factor (P), and criteria for effective collisions.

BOSONANCE

2. Fundamental Concepts & Theoretical Framework

2.1 Rate of a Chemical Reaction

Reactions vary drastically in speed: from instantaneous ionic reactions (such as precipitation of AgCl upon mixing aqueous $\mathrm{AgNO_3}$ and NaCl), to extremely slow transformations (such as diamond converting to graphite, or rusting of iron in air and moisture), to moderate reactions (such as the inversion of cane sugar or hydrolysis of ethyl acetate).

The **rate of a reaction** is defined as the change in concentration of a reactant or product per unit time at constant volume. For a simple general reaction $\mathrm{R} \rightarrow \mathrm{P}$:

$$\text{Rate of disappearance of R} = -\frac{\Delta[\mathrm{R}]}{\Delta t}$$

$$\text{Rate of appearance of P} = +\frac{\Delta[\text{P}]}{\Delta t}$$

Because the concentration of reactant $[\text{R}]$ decreases with time, $\Delta[\text{R}] = [\text{R}]_2 - [\text{R}]_1 < 0$. The negative sign is explicitly prefixed so that the reaction rate remains a strictly positive quantity.

Units of Reaction Rate

Rate has dimensions of $\text{concentration} \times \text{time}^{-1}$. In aqueous or solution systems, units are typically $\text{mol}\,\text{L}^{-1}\,\text{s}^{-1}$ (or $\text{mol}\,\text{L}^{-1}\,\text{min}^{-1}$). For gaseous reactions where concentration is tracked via partial pressures, units are $\text{atm}\,\text{s}^{-1}$ or $\text{bar}\,\text{s}^{-1}$.

2.2 Average Rate vs. Instantaneous Rate

- **Average Rate (r_{av}):** Measured over a finite macroscopic time interval Δt:

$$r_{\text{av}} = -\frac{\Delta[\text{R}]}{\Delta t} = \frac{\Delta[\text{P}]}{\Delta t}$$

  Graphically, r_{av} equals the negative slope of the *secant line* connecting two distinct time points on a $[\text{R}]$ vs. t curve.

- **Instantaneous Rate (r_{inst}):** The rate at a specific moment in time, obtained in the limit as $\Delta t \to 0$:

$$r_{\text{inst}} = \lim_{\Delta t \to 0}\left(-\frac{\Delta[\text{R}]}{\Delta t}\right) = -\frac{\text{d}[\text{R}]}{\text{d}t} = \frac{\text{d}[\text{P}]}{\text{d}t}$$

  Graphically, r_{inst} is equal to the negative slope of the *tangent* drawn to the concentration-time curve at that specific instant t.

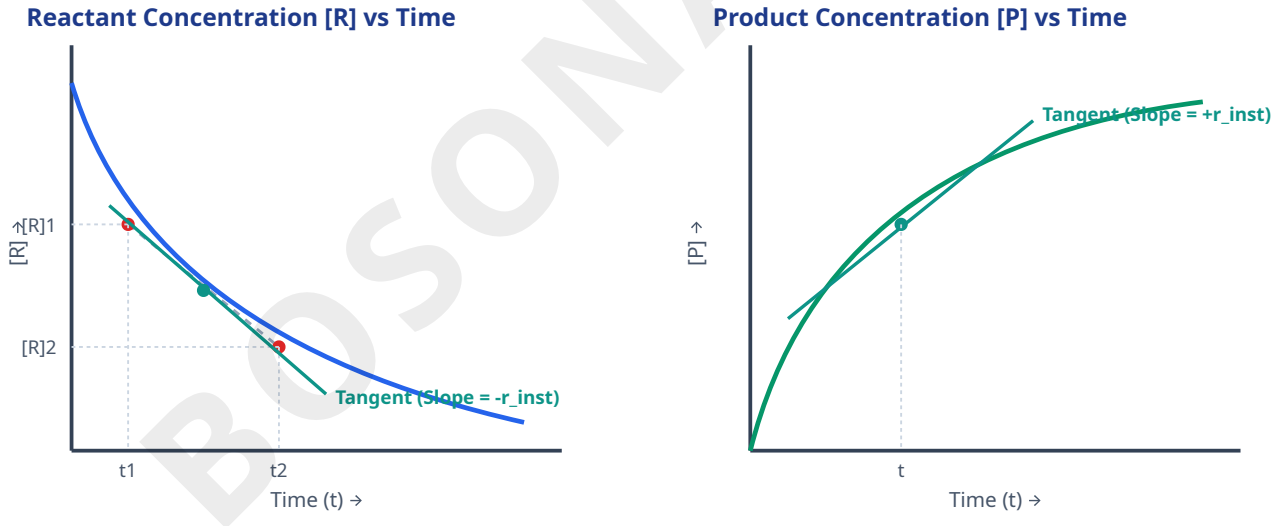


Figure 1: Concentration vs. time curves for reactant disappearance and product appearance, illustrating average rate (secant slope) and instantaneous rate (tangent slope).

2.3 Stoichiometric Expressions of Reaction Rate

When the stoichiometric coefficients of reactants and products are not all unity, the rate of disappearance of each reactant and the rate of appearance of each product must be divided by its respective stoichiometric coefficient so that the overall reaction rate is uniquely defined.

For a general chemical reaction:

$$a\,\text{A} + b\,\text{B} \longrightarrow c\,\text{C} + d\,\text{D}$$

$$\text{Rate of Reaction} = -\frac{1}{a}\frac{\text{d}[\text{A}]}{\text{d}t} = -\frac{1}{b}\frac{\text{d}[\text{B}]}{\text{d}t} = +\frac{1}{c}\frac{\text{d}[\text{C}]}{\text{d}t} = +\frac{1}{d}\frac{\text{d}[\text{D}]}{\text{d}t}$$

BOSONANCE

NEET PITFALL ALERT: RATE OF REACTION VS. RATE OF DISAPPEARANCE/APPEARANCE

Always distinguish between:

- **Rate of disappearance of A:** $-\frac{\mathrm{d[A]}}{\mathrm{d}t}$ (does NOT include the stoichiometric coefficient $1/a$).
- **Rate of appearance of C:** $+\frac{\mathrm{d[C]}}{\mathrm{d}t}$ (does NOT include $1/c$).
- **Rate of the overall reaction:** $-\frac{1}{a}\frac{\mathrm{d[A]}}{\mathrm{d}t} = +\frac{1}{c}\frac{\mathrm{d[C]}}{\mathrm{d}t}$.

2.4 Rate Law, Order of Reaction, and Rate Constant

The **rate law** is an experimentally determined mathematical expression relating the rate of reaction to the molar concentrations of the reactants. For the reaction $a\,\mathrm{A} + b\,\mathrm{B} \rightarrow \text{Products}$:

$$\text{Rate} = k\,[\mathrm{A}]^{x}\,[\mathrm{B}]^{y}$$

- k is the **rate constant** (or specific reaction rate). When $[\mathrm{A}] = [\mathrm{B}] = 1\,\mathrm{mol\,L^{-1}}$, $\text{Rate} = k$. Thus, k represents the reaction velocity when all reactants are at unit molar concentration.
- x is the order with respect to reactant A, and y is the order with respect to B.
- The **overall order of reaction** is $n = x + y$.
- **Crucial Rule:** The powers x and y are determined solely by experiment; they may or may not equal the stoichiometric coefficients a and b. A balanced stoichiometric equation cannot be used to deduce the rate law theoretically!

Units of Rate Constant for an n-th Order Reaction

From the rate law:

$$k = \frac{\text{Rate}}{[\text{Concentration}]^{n}} = \frac{\mathrm{mol\,L^{-1}\,s^{-1}}}{(\mathrm{mol\,L^{-1}})^{n}} = (\mathrm{mol\,L^{-1}})^{1-n}\,\mathrm{s^{-1}} = \mathrm{mol^{1-n}\,L^{n-1}\,s^{-1}}$$

Order (n)	Rate Law Format	SI / Standard Units of Rate Constant (k)	Gaseous Pressure Units
0 (Zero Order)	$\text{Rate} = k$	$\mathrm{mol\,L^{-1}\,s^{-1}}$	$\mathrm{atm\,s^{-1}}$ or $\mathrm{bar\,s^{-1}}$
1 (First Order)	$\text{Rate} = k[\mathrm{A}]$	$\mathrm{s^{-1}}$ (or $\mathrm{min^{-1}}$, $\mathrm{hr^{-1}}$)	$\mathrm{s^{-1}}$
2 (Second Order)	$\text{Rate} = k[\mathrm{A}]^2$	$\mathrm{mol^{-1}\,L\,s^{-1}}$ or $\mathrm{L\,mol^{-1}\,s^{-1}}$	$\mathrm{atm^{-1}\,s^{-1}}$
$3/2$ **(Fractional)**	$\text{Rate} = k[\mathrm{A}]^{3/2}$	$\mathrm{mol^{-1/2}\,L^{1/2}\,s^{-1}}$	$\mathrm{atm^{-1/2}\,s^{-1}}$
n **(General Order)**	$\text{Rate} = k[\mathrm{A}]^{n}$	$\mathrm{mol^{1-n}\,L^{n-1}\,s^{-1}}$	$\mathrm{atm^{1-n}\,s^{-1}}$

2.5 Molecularity vs. Order of Reaction

Chemical reactions can be classified into two fundamental categories:

- **Elementary Reactions:** Chemical reactions that occur in a single collision step without intermediates.
- **Complex Reactions:** Reactions that proceed through a sequence of two or more elementary steps (the *reaction mechanism*).

The **molecularity** of an elementary reaction is defined as the total number of reacting species (atoms, ions, or molecules) that must collide simultaneously in order to bring about the chemical transformation. Simultaneous collisions of more than three molecules are statistically extremely improbable; hence, elementary reactions with molecularity greater than 3 are practically non-existent.

BOSONANCE

Feature	Order of Reaction	Molecularity of Reaction
Definition	Sum of the powers of concentration terms in the experimental rate law.	Number of reacting species that collide simultaneously in an elementary step.
Nature	Purely an **experimental quantity**.	A **theoretical property** derived from the proposed elementary mechanism.
Permissible Values	Can be zero, integer (1, 2, 3), negative, or fractional.	Can only be positive integers: 1 (unimolecular), 2 (bimolecular), 3 (termolecular). Never zero or fractional.
Applicability	Applies to both elementary and complex multi-step reactions.	Has meaning **only for elementary steps**. For complex reactions, molecularity of the overall reaction is meaningless.
Slowest Step Relation	For a complex reaction, overall order is given by the stoichiometry of the slowest step (Rate Determining Step).	Molecularity of the slowest elementary step is identical to the overall order of a simple multi-step sequence without prior equilibria.

Rate Determining Step (RDS) & Reactive Intermediates

In a multi-step complex reaction, the slowest elementary step acts as the **Rate Determining Step (RDS)**, analogous to the slowest runner in a relay race determining the team's total time.

Consider the decomposition of hydrogen peroxide catalyzed by iodide ion in alkaline medium:

$$2\,\mathrm{H_2O_2} \xrightarrow[\text{alkaline medium}]{\mathrm{I^-}} 2\,\mathrm{H_2O} + \mathrm{O_2}$$

The experimentally observed rate law is:

$$\text{Rate} = k\,[\mathrm{H_2O_2}][\mathrm{I^-}]$$

The accepted two-step mechanism is:

1. $\mathrm{H_2O_2} + \mathrm{I^-} \longrightarrow \mathrm{H_2O} + \mathrm{IO^-}$ (Slow, bimolecular, RDS)
2. $\mathrm{H_2O_2} + \mathrm{IO^-} \longrightarrow \mathrm{H_2O} + \mathrm{I^-} + \mathrm{O_2}$ (Fast, bimolecular)

Here, the hypoiodite ion $\mathrm{IO^-}$ is an **intermediate**: it is produced in step 1 and consumed in step 2, so it does not appear in the overall balanced equation. The catalyst $\mathrm{I^-}$ participates in step 1 and is regenerated in step 2.

3. Integrated Rate Laws: Derivations, Graphs, and Half-Lives

While differential rate equations express reaction rate as a function of instantaneous concentration, integrating these equations yields direct functional relationships between measurable concentration $[\mathrm{R}]$, elapsed time t, and rate constant k.

3.1 Zero-Order Reactions

A reaction is of zero order when the rate of reaction is completely independent of the reactant concentration:

$$\text{Rate} = -\frac{\mathrm{d}[\mathrm{R}]}{\mathrm{d}t} = k\,[\mathrm{R}]^0 = k$$
$$\mathrm{d}[\mathrm{R}] = -k\,\mathrm{d}t$$

Integrating both sides between limits $t = 0$ (where $[\mathrm{R}] = [\mathrm{R}]_0$) and time t (where $[\mathrm{R}] = [\mathrm{R}]_t$):

BOSONANCE

$$\int_{[\mathrm{R}]_0}^{[\mathrm{R}]_t} \mathrm{d}[\mathrm{R}] = -k \int_0^t \mathrm{d}t \implies [\mathrm{R}]_t - [\mathrm{R}]_0 = -kt$$

$$[\mathrm{R}]_t = [\mathrm{R}]_0 - kt \quad \iff \quad k = \frac{[\mathrm{R}]_0 - [\mathrm{R}]_t}{t}$$

Zero-Order Half-Life ($t_{1/2}$)

The half-life $t_{1/2}$ is the time required for $[\mathrm{R}]_t$ to drop to $\frac{1}{2}[\mathrm{R}]_0$:

$$\frac{1}{2}[\mathrm{R}]_0 = [\mathrm{R}]_0 - k\,t_{1/2} \implies t_{1/2} = \frac{[\mathrm{R}]_0}{2k}$$

Crucial NEET Takeaway: For a zero-order reaction, $t_{1/2}$ is *directly proportional* to initial concentration: $t_{1/2} \propto [\mathrm{R}]_0$. If initial concentration is doubled, the half-life doubles!

Classic Physical Examples of Zero-Order Reactions

- Decomposition of gaseous ammonia on a hot platinum surface at high pressure:

  $$2\,\mathrm{NH_3(g)} \xrightarrow[1130\ \mathrm{K}]{\mathrm{Pt}} \mathrm{N_2(g)} + 3\,\mathrm{H_2(g)} \qquad \mathrm{Rate} = k\,[\mathrm{NH_3}]^0 = k$$

  Why is it zero-order? At high pressure, all catalytic adsorption sites on the platinum surface become fully saturated. Further increases in $\mathrm{NH_3}$ gas pressure cannot increase the surface concentration of reacting molecules.
- Thermal decomposition of gaseous HI on a gold (Au) surface.
- Enzyme-catalyzed biochemical reactions at substrate saturation.

3.2 First-Order Reactions

In a first-order reaction, the rate is proportional to the first power of reactant concentration:

$$\mathrm{Rate} = -\frac{\mathrm{d}[\mathrm{R}]}{\mathrm{d}t} = k\,[\mathrm{R}]$$

$$\frac{\mathrm{d}[\mathrm{R}]}{[\mathrm{R}]} = -k\,\mathrm{d}t$$

Integrating from $t = 0$ to t:

$$\int_{[\mathrm{R}]_0}^{[\mathrm{R}]_t} \frac{\mathrm{d}[\mathrm{R}]}{[\mathrm{R}]} = -k \int_0^t \mathrm{d}t \implies \ln[\mathrm{R}]_t - \ln[\mathrm{R}]_0 = -kt$$

$$\ln \frac{[\mathrm{R}]_t}{[\mathrm{R}]_0} = -kt \implies [\mathrm{R}]_t = [\mathrm{R}]_0\, e^{-kt}$$

Converting natural log $(\ln)$ to common base-10 log $(\log_{10})$ using $\ln x = 2.303\ \log_{10} x$:

$$k = \frac{2.303}{t} \log_{10}\left(\frac{[\mathrm{R}]_0}{[\mathrm{R}]_t}\right) = \frac{2.303}{t} \log_{10}\left(\frac{a}{a-x}\right)$$

where $a = [\mathrm{R}]_0$ is the initial concentration and x is the concentration of reactant consumed after time t.

BOSONANCE

First-Order Half-Life ($t_{1/2}$)

At $t = t_{1/2}$, $[\mathrm{R}]_t = \frac{[\mathrm{R}]_0}{2}$:

$$k = \frac{2.303}{t_{1/2}} \log_{10}(2) = \frac{2.303 \times 0.3010}{t_{1/2}} = \frac{0.693}{t_{1/2}}$$

$$t_{1/2} = \frac{0.693}{k}$$

Crucial NEET Takeaway: For a first-order reaction, the half-life is strictly *independent* of the initial concentration $[\mathrm{R}]_0$. Whether you start with $10\,\mathrm{M}$ or $0.01\,\mathrm{M}$, it takes the exact same duration for half of the reactant to decompose!

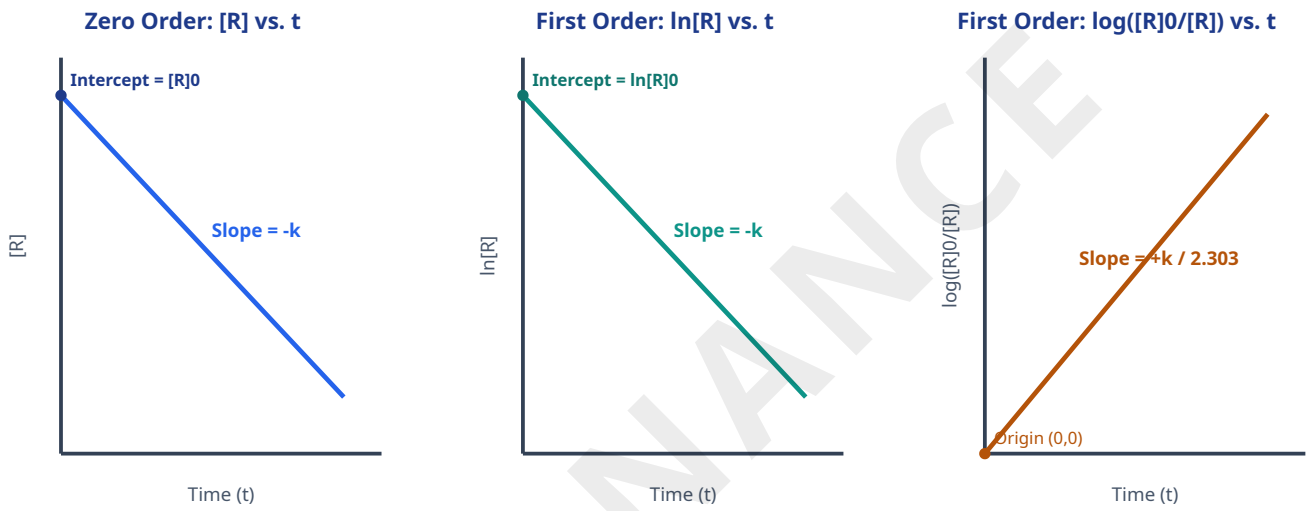


Figure 2: Characteristic linear plots: (A) Zero-order $[\mathrm{R}]$ vs. t; (B) First-order $\ln[\mathrm{R}]$ vs. t; (C) First-order $\log([\mathrm{R}]_0/[\mathrm{R}])$ vs. t.

Key Time Proportions for First-Order Reactions (NEET High-Yield)

- Time for 75% completion ($t_{75\%}$):

$$[\mathrm{R}] = 0.25\,[\mathrm{R}]_0 \implies t_{75\%} = \frac{2.303}{k}\log(4) = \frac{2 \times 2.303 \times 0.3010}{k} = 2 \times t_{1/2}$$

- Time for 87.5% completion ($t_{87.5\%}$): $t_{87.5\%} = 3 \times t_{1/2}$.
- Time for 99.9% completion ($t_{99.9\%}$):

$$[\mathrm{R}] = 0.001\,[\mathrm{R}]_0 = 10^{-3}\,[\mathrm{R}]_0 \implies t_{99.9\%} = \frac{2.303}{k}\log(10^3) = \frac{3 \times 2.303}{k} = \frac{6.909}{k} \approx 10 \times t_{1/2}$$

- Time for 99% vs. 90% completion:

$$t_{99\%} = \frac{2.303}{k}\log(100) = \frac{2 \times 2.303}{k}$$

$$t_{90\%} = \frac{2.303}{k}\log(10) = \frac{2.303}{k} \implies t_{99\%} = 2 \times t_{90\%}$$

3.3 First-Order Gas Phase Reactions

Consider a homogeneous gas-phase reaction where reactant and products are all gases:

$$\mathrm{A(g)} \longrightarrow \mathrm{B(g)} + \mathrm{C(g)}$$

BOSONANCE

Let the initial pressure of pure A at $t = 0$ be p_i. At time t, let x be the decrease in partial pressure of A. According to reaction stoichiometry:

Timeline	p_A	p_B	p_C	Total Pressure (p_t)
At $t = 0$	p_i	0	0	p_i
At time t	$p_\mathrm{i} - x$	x	x	$(p_\mathrm{i} - x) + x + x = p_\mathrm{i} + x$

From total pressure: $p_t = p_\mathrm{i} + x \implies x = p_t - p_\mathrm{i}$.
Substituting x back into the partial pressure of A:

$$p_\mathrm{A} = p_\mathrm{i} - (p_t - p_\mathrm{i}) = 2p_\mathrm{i} - p_t$$
$$k = \frac{2.303}{t} \log_{10}\left(\frac{p_\mathrm{i}}{p_\mathrm{A}}\right) = \frac{2.303}{t} \log_{10}\left(\frac{p_\mathrm{i}}{2p_\mathrm{i} - p_t}\right)$$

3.4 Pseudo-First-Order Reactions

A reaction is termed **pseudo-first-order** when its true molecularity or intrinsic order is higher (e.g., bimolecular/second order), but under actual experimental conditions, it obeys first-order kinetics because one of the reactants is present in large stoichiometric excess.

1. Acid Hydrolysis of Ethyl Acetate

$$\mathrm{CH_3COOC_2H_5} + \mathrm{H_2O} \xrightarrow{\mathrm{H^+}} \mathrm{CH_3COOH} + \mathrm{C_2H_5OH}$$

The true rate equation is $\mathrm{Rate} = k' \left[\mathrm{CH_3COOC_2H_5}\right]\left[\mathrm{H_2O}\right]$. When $0.01\,\mathrm{mol}$ of ethyl acetate is reacted in $10\,\mathrm{mol}$ of water, the water concentration decreases only from $10\,\mathrm{mol}$ to $9.99\,\mathrm{mol}$—an insignificant change. Hence, $\left[\mathrm{H_2O}\right]$ remains effectively constant:

$$\mathrm{Rate} = k' \left[\mathrm{H_2O}\right] \left[\mathrm{CH_3COOC_2H_5}\right] = k \left[\mathrm{CH_3COOC_2H_5}\right] \qquad (\text{where } k = k' \left[\mathrm{H_2O}\right])$$

2. Inversion of Cane Sugar (Sucrose)

$$\mathrm{C_{12}H_{22}O_{11}} + \mathrm{H_2O} \xrightarrow{\mathrm{H^+}} \underset{\text{D-glucose}}{\mathrm{C_6H_{12}O_6}} + \underset{\text{D-fructose}}{\mathrm{C_6H_{12}O_6}}$$
$$\mathrm{Rate} = k \left[\mathrm{C_{12}H_{22}O_{11}}\right]$$

4. Temperature Dependence, Arrhenius Equation, and Catalysis

4.1 Temperature Coefficient (Q_{10})

For most chemical reactions, the rate of reaction increases exponentially with temperature. As a general empirical rule, **a rise in temperature of $10\,^\circ\mathrm{C}$ (or $10\,\mathrm{K}$) nearly doubles or triples the rate constant**:

$$\text{Temperature Coefficient } (\mu \text{ or } Q_{10}) = \frac{k_{T+10}}{k_T} \approx 2 \text{ to } 3$$

4.2 The Arrhenius Equation

Svante Arrhenius provided the physical interpretation of temperature dependence based on the formation of an energized intermediate known as the **activated complex**:

BOSONANCE

$$k = A\, e^{-E_a/RT}$$

- A: Arrhenius factor, frequency factor, or pre-exponential factor (has identical units to k).
- E_a: **Activation energy** in $\mathrm{J\,mol^{-1}}$ or $\mathrm{kJ\,mol^{-1}}$. It represents the minimum extra energy that reactant molecules must absorb to reach the transition state.
- R: Universal gas constant ($8.314\,\mathrm{J\,K^{-1}\,mol^{-1}}$).
- T: Absolute temperature in Kelvin (K).
- The Boltzmann factor $e^{-E_a/RT}$ represents the **fraction of molecules** having kinetic energy equal to or exceeding the activation energy E_a.

Logarithmic and Two-Temperature Forms

Taking natural logarithm:

$$\ln k = -\frac{E_a}{R}\left(\frac{1}{T}\right) + \ln A$$

Comparing with the equation of a straight line ($y = mx + c$), a plot of $\ln k$ against $1/T$ yields a straight line with:

- $\text{Slope} = -\frac{E_a}{R}$
- $\text{Intercept} = \ln A$

In base-10 logarithms between two temperatures T_1 and T_2 (with rate constants k_1 and k_2):

$$\log_{10}\left(\frac{k_2}{k_1}\right) = \frac{E_a}{2.303\,R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right] = \frac{E_a}{2.303\,R}\left[\frac{T_2 - T_1}{T_1 T_2}\right]$$

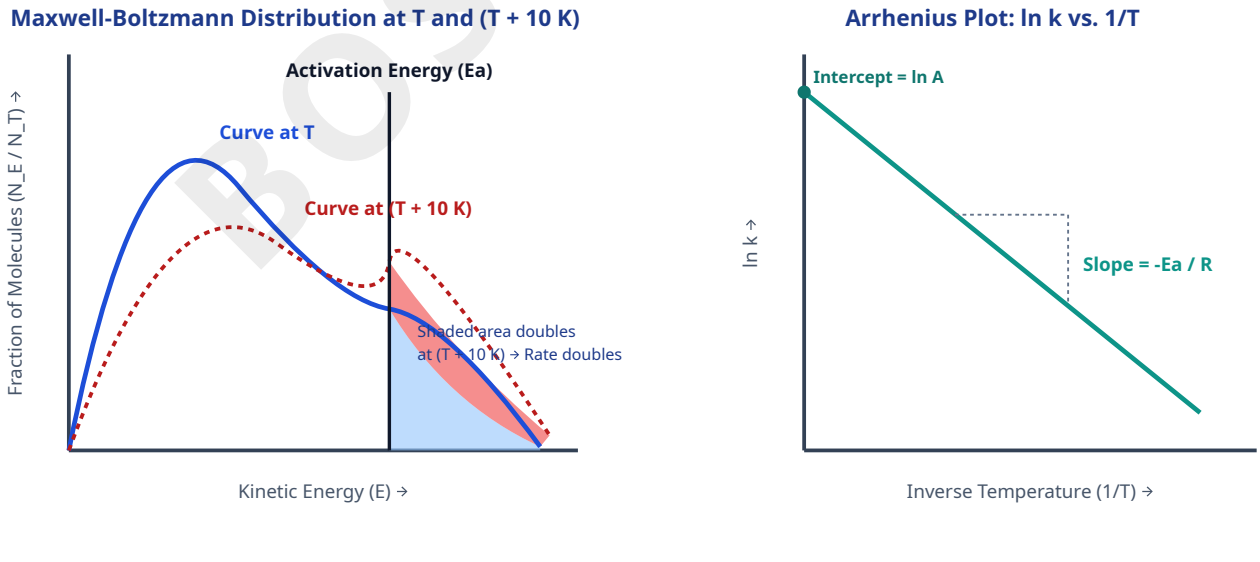


Figure 3: (Left) Maxwell-Boltzmann distribution curves showing why a $10\,\mathrm{K}$ temperature rise roughly doubles the fraction of molecules possessing energy $\geq E_a$; (Right) Arrhenius plot of $\ln k$ against $1/T$.

4.3 Effect of Catalyst: Intermediate Complex Theory

A **catalyst** is a substance that accelerates the rate of a chemical reaction without itself undergoing any permanent chemical change. According to the *intermediate complex theory*, the catalyst combines

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temporarily with reactant molecules to form an unstable intermediate complex, which then decomposes into products and regenerates the catalyst intact.

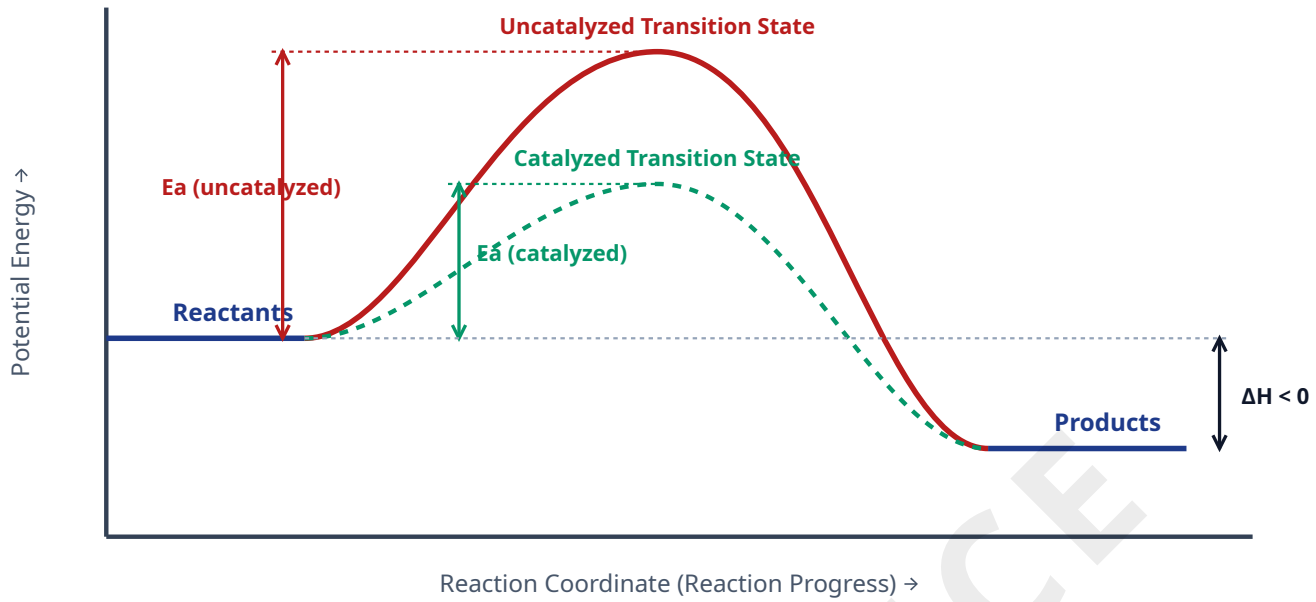


Figure 4: Potential energy profile of an exothermic reaction showing the uncatalyzed pathway versus the lower activation energy pathway provided by a catalyst. Note that reactant energy, product energy, and ΔH remain identical.

NEET MASTER RULES: WHAT A CATALYST DOES AND DOES NOT DO

- **Does:** Provides an alternative mechanism with a lower potential energy barrier (lowers E_a for both forward and reverse reactions by the exact same amount).
- **Does:** Increases both forward rate constant (k_f) and backward rate constant (k_b) to the exact same extent, thereby reducing the time taken to attain equilibrium.
- **Does NOT:** Alter the equilibrium constant ($K_{eq} = k_f/k_b$ remains unchanged).
- **Does NOT:** Alter standard Gibbs free energy (ΔG°) or enthalpy (ΔH) of the reaction.
- **Does NOT:** Make a non-spontaneous reaction ($\Delta G > 0$) spontaneous; it only accelerates thermodynamically feasible processes.

4.4 Collision Theory of Chemical Reactions

Developed by Max Trautz and William Lewis (1916–18) based on the kinetic theory of gases, Collision Theory treats molecules as hard rigid spheres. A chemical reaction occurs only when reactant molecules physically collide. However, only a tiny fraction of total collisions lead to chemical reaction.

The Two Necessary Barriers for an Effective Collision

1. **Energy Barrier (Threshold Energy):** Colliding molecules must possess a minimum amount of energy called **Threshold Energy**.

$$\text{Threshold Energy} = E_a + \text{Initial kinetic energy possessed by reacting molecules}$$

   Collisions involving kinetic energy below E_a are elastic: the molecules simply rebound without reacting.
2. **Orientation Barrier (Steric Factor P):** Even if molecules possess sufficient kinetic energy, bond rearrangement cannot occur unless reacting atoms collide in the correct spatial orientation.

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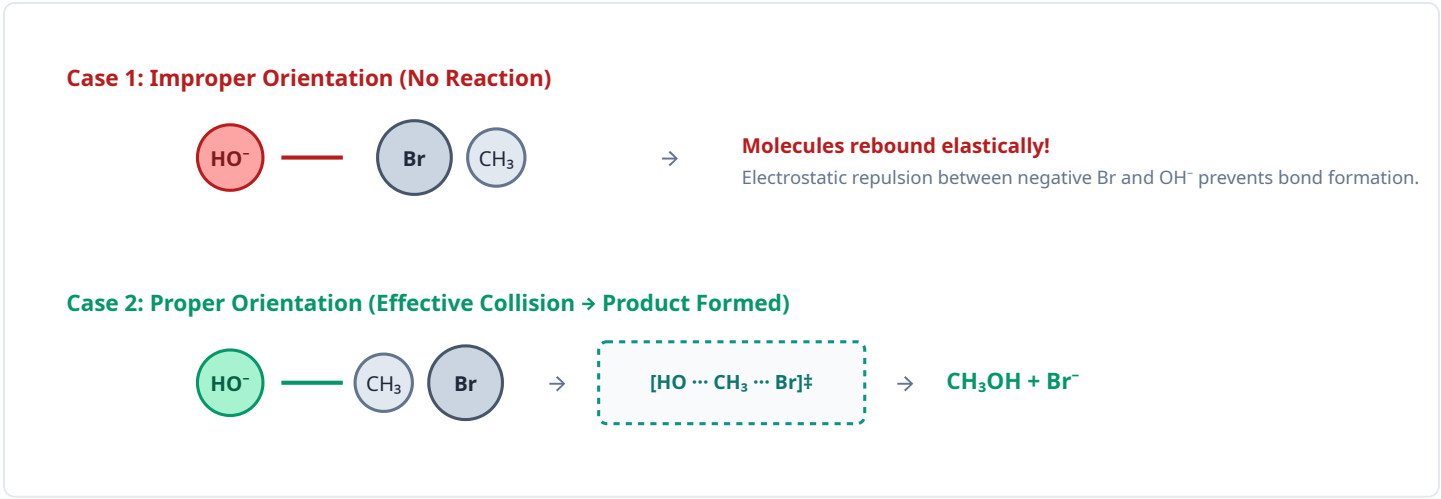


Figure 5: Collision theory model of molecular orientation illustrated for the bimolecular nucleophilic substitution of bromomethane with hydroxide ion.

To incorporate both requirements, the modified collision rate equation is formulated as:

$$\text{Rate} = P\, Z_{\text{AB}}\, e^{-E_{\text{a}}/RT}$$

- Z_{AB}: Collision frequency, defined as the number of collisions per second per unit volume of the reaction mixture between species A and B.
- P: Steric factor or probability factor ($P \leq 1$), which accounts for the proper orientation of reacting molecules.
- Arrhenius pre-exponential factor is related as: $A = P\, Z_{\text{AB}}$.
- **Drawback of Collision Theory:** Treats atoms/molecules as hard structureless spheres and completely ignores their internal degrees of freedom, vibrations, and rotational electron cloud distortions.

5. Master Summary of Kinetic Formulations

Kinetic Property	Zero Order ($n = 0$)	First Order ($n = 1$)	n-th Order ($n \neq 1$)
Differential Rate Law	$-\frac{\text{d}[\text{R}]}{\text{d}t} = k$	$-\frac{\text{d}[\text{R}]}{\text{d}t} = k[\text{R}]$	$-\frac{\text{d}[\text{R}]}{\text{d}t} = k[\text{R}]^n$
Integrated Rate Law	$[\text{R}]_t = [\text{R}]_0 - kt$	$[\text{R}]_t = [\text{R}]_0\, e^{-kt}$ or $k = \frac{2.303}{t} \log \frac{[\text{R}]_0}{[\text{R}]_t}$	$\frac{1}{[\text{R}]_t^{n-1}} - \frac{1}{[\text{R}]_0^{n-1}} = (n-1)kt$
Linear Plot (y vs x)	$[\text{R}]$ vs t (Slope $= -k$)	$\ln[\text{R}]$ vs t (Slope $= -k$)	$\frac{1}{[\text{R}]^{n-1}}$ vs t (Slope $= (n-1)k$)
Half-Life ($t_{1/2}$)	$t_{1/2} = \frac{[\text{R}]_0}{2k}$	$t_{1/2} = \frac{\ln 2}{k} = \frac{0.693}{k}$	$t_{1/2} \propto \frac{1}{[\text{R}]_0^{n-1}}$
Dependence of $t_{1/2}$ on $[\text{R}]_0$	$t_{1/2} \propto [\text{R}]_0^1$	Independent of $[\text{R}]_0$ ($t_{1/2} \propto [\text{R}]_0^0$)	$t_{1/2} \propto [\text{R}]_0^{1-n}$
Units of Rate Constant (k)	$\text{mol}\,\text{L}^{-1}\,\text{s}^{-1}$	s^{-1}	$\text{mol}^{1-n}\,\text{L}^{n-1}\,\text{s}^{-1}$

BOSONANCE

6. Representative Worked Examples (Step-by-Step)

Worked Example 1: Stoichiometric Rates of Reaction and Appearance

Problem: In the high-temperature decomposition of dinitrogen pentoxide in carbon tetrachloride:

$$2\,\mathrm{N_2O_5(g)} \longrightarrow 4\,\mathrm{NO_2(g)} + \mathrm{O_2(g)}$$

The concentration of $\mathrm{N_2O_5}$ drops from $2.33\,\mathrm{mol\,L^{-1}}$ to $2.08\,\mathrm{mol\,L^{-1}}$ in $184\,\mathrm{minutes}$. Calculate:

1. The average rate of the reaction in $\mathrm{mol\,L^{-1}\,min^{-1}}$ and $\mathrm{mol\,L^{-1}\,s^{-1}}$.
2. The rate of production of $\mathrm{NO_2}$ during this period.

Step-by-Step Solution:

Step 1: Express the overall average reaction rate from stoichiometry:

$$\text{Rate} = -\frac{1}{2}\frac{\Delta[\mathrm{N_2O_5}]}{\Delta t} = \frac{1}{4}\frac{\Delta[\mathrm{NO_2}]}{\Delta t} = \frac{\Delta[\mathrm{O_2}]}{\Delta t}$$

Step 2: Substitute concentration changes for $\mathrm{N_2O_5}$:

$$\Delta[\mathrm{N_2O_5}] = 2.08 - 2.33 = -0.25\,\mathrm{mol\,L^{-1}}$$

$$\Delta t = 184\,\mathrm{min}$$

$$\text{Rate} = -\frac{1}{2}\left(\frac{-0.25\,\mathrm{mol\,L^{-1}}}{184\,\mathrm{min}}\right) = \frac{0.25}{368} = 6.79 \times 10^{-4}\,\mathrm{mol\,L^{-1}\,min^{-1}}$$

In seconds:

$$\text{Rate} = \frac{6.79 \times 10^{-4}\,\mathrm{mol\,L^{-1}\,min^{-1}}}{60\,\mathrm{s\,min^{-1}}} = 1.13 \times 10^{-5}\,\mathrm{mol\,L^{-1}\,s^{-1}}$$

Step 3: Calculate rate of formation of $\mathrm{NO_2}$:

$$\frac{\Delta[\mathrm{NO_2}]}{\Delta t} = 4 \times \text{Rate} = 4 \times (6.79 \times 10^{-4}\,\mathrm{mol\,L^{-1}\,min^{-1}}) = 2.72 \times 10^{-3}\,\mathrm{mol\,L^{-1}\,min^{-1}}$$

BOSONANCE

Worked Example 2: Deducing Rate Law from Initial Rate Data

Problem: For the reaction $2\,\mathrm{A} + \mathrm{B} \rightarrow \mathrm{C} + \mathrm{D}$, the following initial rates were obtained at $300\,\mathrm{K}$:

Run	$[\mathrm{A}]_0 / \mathrm{mol\,L^{-1}}$	$[\mathrm{B}]_0 / \mathrm{mol\,L^{-1}}$	Initial Rate of Formation of $\mathrm{D} / \mathrm{mol\,L^{-1}\,min^{-1}}$
I	0.10	0.10	6.0×10^{-3}
II	0.30	0.20	7.2×10^{-2}
III	0.30	0.40	2.88×10^{-1}
IV	0.40	0.10	2.40×10^{-2}

Determine the order with respect to A and B, write the rate law, and calculate the rate constant k.

Step-by-Step Solution:

Let the rate law be: $\mathrm{Rate} = k\,[\mathrm{A}]^x\,[\mathrm{B}]^y$.

Step 1: Determine order with respect to A (holding $[\mathrm{B}]$ constant):

Compare Run IV and Run I (where $[\mathrm{B}] = 0.10\,\mathrm{M}$):

$$\frac{\text{Rate IV}}{\text{Rate I}} = \frac{2.40 \times 10^{-2}}{6.0 \times 10^{-3}} = 4.0$$

$$\frac{[\mathrm{A}]_{\mathrm{IV}}^x}{[\mathrm{A}]_{\mathrm{I}}^x} = \left(\frac{0.40}{0.10}\right)^x = (4)^x \implies 4^x = 4 \implies x = 1$$

The reaction is **first order with respect to A**.

Step 2: Determine order with respect to B (holding $[\mathrm{A}]$ constant):

Compare Run III and Run II (where $[\mathrm{A}] = 0.30\,\mathrm{M}$):

$$\frac{\text{Rate III}}{\text{Rate II}} = \frac{2.88 \times 10^{-1}}{7.2 \times 10^{-2}} = \frac{0.288}{0.072} = 4.0$$

$$\frac{[\mathrm{B}]_{\mathrm{III}}^y}{[\mathrm{B}]_{\mathrm{II}}^y} = \left(\frac{0.40}{0.20}\right)^y = (2)^y \implies 2^y = 4 \implies y = 2$$

The reaction is **second order with respect to B**.

Step 3: Write overall rate law and calculate k:

$$\mathrm{Rate} = k\,[\mathrm{A}]\,[\mathrm{B}]^2 \qquad (\text{Overall order } = 1 + 2 = 3)$$

Using data from Run I:

$$6.0 \times 10^{-3}\,\mathrm{mol\,L^{-1}\,min^{-1}} = k\,(0.10\,\mathrm{mol\,L^{-1}})\,(0.10\,\mathrm{mol\,L^{-1}})^2 = k \times 10^{-3}\,\mathrm{mol^3\,L^{-3}}$$

$$k = \frac{6.0 \times 10^{-3}}{1.0 \times 10^{-3}} = 6.0\,\mathrm{mol^{-2}\,L^2\,min^{-1}}$$

BOSONANCE

Worked Example 3: Gas-Phase First-Order Decomposition with Pressure Monitoring

Problem: Thermal decomposition of gaseous $\mathrm{N_2O_5}$ was monitored at constant volume at $318\,\mathrm{K}$:

$$2\,\mathrm{N_2O_5(g)} \longrightarrow 2\,\mathrm{N_2O_4(g)} + \mathrm{O_2(g)}$$

The initial total pressure at $t = 0$ was $0.500\,\mathrm{atm}$. After $100\,\mathrm{s}$, the total pressure rose to $0.512\,\mathrm{atm}$. Calculate the first-order rate constant k.

Step-by-Step Solution:

Let the initial pressure of $\mathrm{N_2O_5}$ be $p_0 = 0.500\,\mathrm{atm}$. Let $2x$ be the pressure of $\mathrm{N_2O_5}$ that decomposes in $100\,\mathrm{s}$:

$$p_{\mathrm{N_2O_5}} = 0.500 - 2x$$

$$p_{\mathrm{N_2O_4}} = 2x, \qquad p_{\mathrm{O_2}} = x$$

$$p_t = p_{\mathrm{N_2O_5}} + p_{\mathrm{N_2O_4}} + p_{\mathrm{O_2}} = (0.500 - 2x) + 2x + x = 0.500 + x$$

Given $p_t = 0.512\,\mathrm{atm}$:

$$x = 0.512 - 0.500 = 0.012\,\mathrm{atm}$$

The remaining partial pressure of $\mathrm{N_2O_5}$ is:

$$p_{\mathrm{N_2O_5}} = 0.500 - 2(0.012) = 0.500 - 0.024 = 0.476\,\mathrm{atm}$$

For a first-order reaction:

$$k = \frac{2.303}{t}\log_{10}\left(\frac{p_0}{p_{\mathrm{N_2O_5}}}\right) = \frac{2.303}{100}\log_{10}\left(\frac{0.500}{0.476}\right) = \frac{2.303}{100}\log_{10}(1.0504)$$

$$\log_{10}(1.0504) \approx 0.02136$$

$$k = \frac{2.303}{100} \times 0.02136 = 4.92 \times 10^{-4}\,\mathrm{s^{-1}}$$

Worked Example 4: Arrhenius Activation Energy and Two-Temperature Calculation

Problem: The rate constant for the decomposition of ethyl iodide ($\mathrm{C_2H_5I} \rightarrow \mathrm{C_2H_4} + \mathrm{HI}$) at $600\,\mathrm{K}$ is $1.60 \times 10^{-5}\,\mathrm{s^{-1}}$. Its activation energy is $209\,\mathrm{kJ\,mol^{-1}}$. Calculate the rate constant at $700\,\mathrm{K}$ ($R = 8.314\,\mathrm{J\,K^{-1}\,mol^{-1}}$).

Step-by-Step Solution:

Using the Arrhenius two-temperature relation:

$$\log_{10}\left(\frac{k_2}{k_1}\right) = \frac{E_\mathrm{a}}{2.303\,R}\left[\frac{T_2 - T_1}{T_1 T_2}\right]$$

Substitute known values: $E_\mathrm{a} = 209000\,\mathrm{J\,mol^{-1}}$, $T_1 = 600\,\mathrm{K}$, $T_2 = 700\,\mathrm{K}$:

$$\frac{E_\mathrm{a}}{2.303\,R} = \frac{209000}{2.303 \times 8.314} = \frac{209000}{19.147} \approx 10915.5\,\mathrm{K}$$

$$\frac{T_2 - T_1}{T_1 T_2} = \frac{700 - 600}{600 \times 700} = \frac{100}{420000} = \frac{1}{4200}\,\mathrm{K^{-1}}$$

$$\log_{10}\left(\frac{k_2}{k_1}\right) = \frac{10915.5}{4200} = 2.599$$

Taking antilog:

$$\frac{k_2}{k_1} = 10^{2.599} = 397.2$$

$$k_2 = 397.2 \times k_1 = 397.2 \times (1.60 \times 10^{-5}\,\mathrm{s^{-1}}) = 6.36 \times 10^{-3}\,\mathrm{s^{-1}}$$

BOSONANCE

7. NEET UG Practice Assessment (Exam-Aligned Questions & Solutions)

The following original questions are formatted strictly according to the latest NEET UG examination pattern, including Single Correct MCQs, Assertion-Reason pairs, Statement-I / Statement-II analyses, and Matching questions.

Practice Question 1 Single Correct MCQ

For the gaseous reaction $2\,\mathrm{A} + \mathrm{B} \longrightarrow 3\,\mathrm{C}$, the rate of formation of C is $0.060\,\mathrm{mol\,L^{-1}\,s^{-1}}$. The rate of disappearance of A and the overall rate of reaction are, respectively:

(A) $0.040\,\mathrm{mol\,L^{-1}\,s^{-1}}$ and $0.020\,\mathrm{mol\,L^{-1}\,s^{-1}}$

(B) $0.090\,\mathrm{mol\,L^{-1}\,s^{-1}}$ and $0.060\,\mathrm{mol\,L^{-1}\,s^{-1}}$

(C) $0.040\,\mathrm{mol\,L^{-1}\,s^{-1}}$ and $0.060\,\mathrm{mol\,L^{-1}\,s^{-1}}$

(D) $0.020\,\mathrm{mol\,L^{-1}\,s^{-1}}$ and $0.040\,\mathrm{mol\,L^{-1}\,s^{-1}}$

Correct Answer: (A)

Explanation: According to reaction stoichiometry:

$$\text{Overall Rate} = -\frac{1}{2}\frac{\mathrm{d[A]}}{\mathrm{d}t} = -\frac{\mathrm{d[B]}}{\mathrm{d}t} = +\frac{1}{3}\frac{\mathrm{d[C]}}{\mathrm{d}t}$$

Given $\frac{\mathrm{d[C]}}{\mathrm{d}t} = 0.060\,\mathrm{mol\,L^{-1}\,s^{-1}}$:

$$\text{Overall Rate} = \frac{1}{3}(0.060) = 0.020\,\mathrm{mol\,L^{-1}\,s^{-1}}$$

The rate of disappearance of A is $-\frac{\mathrm{d[A]}}{\mathrm{d}t}$:

$$-\frac{\mathrm{d[A]}}{\mathrm{d}t} = 2 \times \text{Overall Rate} = 2 \times 0.020 = 0.040\,\mathrm{mol\,L^{-1}\,s^{-1}}$$

BOSONANCE

Practice Question 2 Single Correct MCQ

The rate constant of a certain reaction is given as $k = 4.5 \times 10^{-3}\,\text{mol}^{-1/2}\,\text{L}^{1/2}\,\text{s}^{-1}$. The overall order of this reaction is:

(A) $1/2$

(B) $3/2$

(C) 2

(D) $-1/2$

Correct Answer: (B)

Explanation: The general dimensional formula for rate constant is:

$$\text{Units of } k = \text{mol}^{1-n}\,\text{L}^{n-1}\,\text{s}^{-1}$$

Equating exponents of L:

$$n - 1 = \frac{1}{2} \implies n = 1 + \frac{1}{2} = \frac{3}{2}$$

Verifying with exponent of mol: $1 - n = 1 - 3/2 = -1/2$. Hence, order is $3/2$.

Practice Question 3 Single Correct MCQ

A first-order reaction is 20% complete in $10\,\text{minutes}$. How much time will it take to reach 80% completion? [Given: $\log_{10}(2) = 0.3010$]

(A) $40.0\,\text{min}$

(B) $72.1\,\text{min}$

(C) $62.2\,\text{min}$

(D) $80.0\,\text{min}$

Correct Answer: (B)

Explanation: For a first order reaction, $k = \frac{2.303}{t}\log\left(\frac{a}{a-x}\right)$.

At 20% completion: $a - x = 0.80\,a$, $t_1 = 10\,\text{min}$:

$$k = \frac{2.303}{10}\log\left(\frac{1}{0.80}\right) = \frac{2.303}{10}\log(1.25)$$

$$\log(1.25) = \log(10/8) = 1 - 3\log 2 = 1 - 0.9030 = 0.0970$$

At 80% completion: $a - x = 0.20\,a$:

$$k = \frac{2.303}{t_2}\log\left(\frac{1}{0.20}\right) = \frac{2.303}{t_2}\log(5)$$

$$\log(5) = 1 - \log 2 = 1 - 0.3010 = 0.6990$$

Dividing the two equations:

$$\frac{t_2}{10} = \frac{\log 5}{\log 1.25} = \frac{0.6990}{0.0970} \approx 7.206 \implies t_2 = 72.06 \approx 72.1\,\text{min}$$

BOSONANCE

Practice Question 4 Single Correct MCQ

A plot of $t_{1/2}$ versus initial concentration $[\mathrm{R}]_0$ for a chemical reaction yields a horizontal straight line parallel to the concentration axis. When the initial concentration is increased four-fold, the rate of reaction increases by a factor of:

(A) 1 (remains unchanged)

(B) 4

(C) 16

(D) 2

Correct Answer: (B)
Explanation: If $t_{1/2}$ is constant and independent of $[\mathrm{R}]_0$, the reaction is strictly **first order** ($n = 1$).

For a first-order reaction: $\mathrm{Rate} = k[\mathrm{R}]^1$. When $[\mathrm{R}]_0$ is increased 4 times, the rate increases by $4^1 = 4$ times.

Practice Question 5 Assertion & Reason (NEET Format)

Given below are two statements labeled as Assertion (A) and Reason (R):

Assertion (A): The molecularity of an elementary reaction can never be zero, negative, or fractional.

Reason (R): Molecularity represents the number of reacting molecules participating in simultaneous collisions in an elementary step, and colliding species must be whole physical entities.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

Correct Answer: (A)
Explanation: Molecularity is defined strictly as the integer count of molecules undergoing simultaneous collision in an elementary step. One cannot have zero molecules colliding to form a product, nor can a fraction of a molecule undergo collision. Hence, molecularity must be a non-zero positive integer (1, 2, or 3). Reason (R) correctly explains Assertion (A).

BOSONANCE

Practice Question 6

Statement I & Statement II (NEET Format)

Examine the two statements given below:

Statement I: A positive catalyst increases the rate of both the forward and backward reactions by the same extent without altering ΔG or the equilibrium constant K_{eq}.

Statement II: A catalyst accelerates the rate by lowering the activation energy through the provision of an alternate reaction mechanism involving an intermediate complex.

(A) Both Statement I and Statement II are correct.

(B) Both Statement I and Statement II are incorrect.

(C) Statement I is correct but Statement II is incorrect.

(D) Statement I is incorrect but Statement II is correct.

Correct Answer: (A)

Explanation: Both statements are completely true scientific facts from the chapter. A catalyst reduces E_{a} equally in both directions, so both k_{f} and k_{b} increase by the same factor, leaving $K_{\text{eq}} = k_{\text{f}}/k_{\text{b}}$ unchanged. Thermodynamic parameters (ΔG, ΔH) are state functions and do not depend on the catalytic pathway.

Practice Question 7

Single Correct MCQ

The rate of a certain chemical reaction doubles when the temperature is raised from $298\,\text{K}$ to $308\,\text{K}$. The activation energy (E_{a}) of the reaction is approximately: [Take $R = 8.314\,\text{J}\,\text{K}^{-1}\,\text{mol}^{-1}$, $\log_{10}(2) = 0.3010$]

(A) $52.9\,\text{kJ}\,\text{mol}^{-1}$

(B) $105.8\,\text{kJ}\,\text{mol}^{-1}$

(C) $26.5\,\text{kJ}\,\text{mol}^{-1}$

(D) $83.1\,\text{kJ}\,\text{mol}^{-1}$

Correct Answer: (A)

Explanation: Using the Arrhenius equation:

$$\log_{10}\left(\frac{k_2}{k_1}\right) = \frac{E_{\text{a}}}{2.303\,R}\left[\frac{T_2 - T_1}{T_1 T_2}\right]$$

$$\log(2) = 0.3010 = \frac{E_{\text{a}}}{2.303 \times 8.314}\left[\frac{308 - 298}{298 \times 308}\right]$$

$$0.3010 = \frac{E_{\text{a}}}{19.147}\left[\frac{10}{91784}\right]$$

$$E_{\text{a}} = \frac{0.3010 \times 19.147 \times 91784}{10} = 52897\,\text{J}\,\text{mol}^{-1} \approx 52.9\,\text{kJ}\,\text{mol}^{-1}$$

BOSONANCE

Practice Question 8 Single Correct MCQ

For a first-order gas-phase decomposition: $2\,\mathrm{A(g)} \longrightarrow 2\,\mathrm{B(g)} + \mathrm{C(g)}$, the initial pressure of A is p_0. After time t, the total pressure becomes p_t. The correct expression for the rate constant k is:

(A) $k = \frac{2.303}{t}\log_{10}\left(\frac{p_0}{2p_0 - p_t}\right)$

(B) $k = \frac{2.303}{t}\log_{10}\left(\frac{p_0}{3p_0 - 2p_t}\right)$

(C) $k = \frac{2.303}{t}\log_{10}\left(\frac{2p_0}{3p_0 - p_t}\right)$

(D) $k = \frac{2.303}{t}\log_{10}\left(\frac{p_0}{p_t - p_0}\right)$

Correct Answer: (B)

Explanation: Let the decrease in pressure of A be $2x$:

$$p_{\mathrm{A}} = p_0 - 2x, \quad p_{\mathrm{B}} = 2x, \quad p_{\mathrm{C}} = x$$
$$p_t = (p_0 - 2x) + 2x + x = p_0 + x \implies x = p_t - p_0$$

Substitute x into p_{A}:

$$p_{\mathrm{A}} = p_0 - 2(p_t - p_0) = 3p_0 - 2p_t$$

Therefore, $k = \frac{2.303}{t}\log_{10}\left(\frac{p_0}{p_{\mathrm{A}}}\right) = \frac{2.303}{t}\log_{10}\left(\frac{p_0}{3p_0 - 2p_t}\right)$.

Practice Question 9 Single Correct MCQ

A radioactive isotope $^{90}\mathrm{Sr}$ has a half-life of $28.1\ \text{years}$. If $1.0\ \mu\text{g}$ of this isotope is absorbed in bone tissue, the mass remaining after $56.2\ \text{years}$ is:

(A) $0.50\ \mu\text{g}$

(B) $0.25\ \mu\text{g}$

(C) $0.125\ \mu\text{g}$

(D) $0.75\ \mu\text{g}$

Correct Answer: (B)

Explanation: All radioactive decays follow first-order kinetics. The number of elapsed half-lives is:

$$n = \frac{t}{t_{1/2}} = \frac{56.2\,\text{yr}}{28.1\,\text{yr}} = 2$$

$$N = N_0\left(\frac{1}{2}\right)^n = 1.0\,\mu\text{g} \times \left(\frac{1}{2}\right)^2 = 0.25\,\mu\text{g}$$

BOSONANCE

Practice Question 10

Match the Column (NEET Format)

Match the reaction types in **Column-I** with their characteristic kinetic features in **Column-II**:

Column-I	Column-II
(P) Zero-order reaction	(1) Hydrolysis of ethyl acetate in large excess of water
(Q) Pseudo-first-order reaction	(2) Rate constant has unit $\mathrm{s^{-1}}$
(R) First-order reaction	(3) $t_{1/2}$ is directly proportional to initial concentration
(S) Second-order reaction	(4) Rate constant has unit $\mathrm{L\,mol^{-1}\,s^{-1}}$

(A) P-(3), Q-(1), R-(2), S-(4)

(B) P-(1), Q-(3), R-(4), S-(2)

(C) P-(3), Q-(2), R-(1), S-(4)

(D) P-(4), Q-(1), R-(2), S-(3)

Correct Answer: (A)

Explanation:

- (P) For zero order: $t_{1/2} = [\mathrm{R}]_0/(2k) \propto [\mathrm{R}]_0$ → matches (3).
- (Q) Hydrolysis of ethyl acetate in excess water is a pseudo-first-order reaction → matches (1).
- (R) First-order rate constant has unit $\mathrm{s^{-1}}$ → matches (2).
- (S) Second-order rate constant has unit $\mathrm{L\,mol^{-1}\,s^{-1}}$ → matches (4).

Practice Question 11

Single Correct MCQ

The rate equation for the decomposition of a substance is: $\mathrm{Rate} = k\,[\mathrm{A}]^2$. If the initial concentration of A is tripled, the rate of decomposition of A will:

(A) Increase 3 times

(B) Increase 6 times

(C) Increase 9 times

(D) Remain unchanged

Correct Answer: (C)

Explanation: $\mathrm{Rate}_1 = k[\mathrm{A}]^2$. When $[\mathrm{A}]' = 3[\mathrm{A}]$:

$$\mathrm{Rate}_2 = k(3[\mathrm{A}])^2 = 9\,k[\mathrm{A}]^2 = 9 \times \mathrm{Rate}_1$$

The rate increases 9-fold.

BOSONANCE

Practice Question 12 Single Correct MCQ

According to Collision Theory, the rate of a bimolecular elementary reaction $\mathrm{A} + \mathrm{B} \rightarrow \mathrm{Products}$ is given by $\mathrm{Rate} = P\, Z_{\mathrm{AB}}\, e^{-E_a/RT}$. In this expression, the steric factor P accounts for:

(A) The fraction of molecules having energy greater than threshold energy.

(B) The proper three-dimensional orientation of colliding molecules.

(C) The total number of collisions taking place per second per unit volume.

(D) The mass of the colliding hard spheres.

Correct Answer: (B)

Explanation: In collision theory, Z_{AB} is collision frequency, $e^{-E_a/RT}$ is the Boltzmann fraction possessing required activation energy, and the steric/probability factor P accounts for the necessity of **proper molecular orientation** during collision to facilitate bond cleavages and bond formations.

8. Common Pitfalls & NEET High-Yield Revision Checklist

FREQUENT STUDENT MISTAKES TO AVOID

- **Confusing Rate of Disappearance with Overall Rate:** Remember that in $2\mathrm{A} \rightarrow \mathrm{Products}$, $-\frac{\mathrm{d[A]}}{\mathrm{d}t} = 2 \times \mathrm{Rate\ of\ Reaction}$.
- **Predicting Rate Law from Stoichiometry:** Stoichiometric coefficients tell you nothing about the rate law unless you are explicitly told the reaction is *elementary*.
- **Unit Confusion in Rate Constants:** Always use $\mathrm{mol}^{1-n}\,\mathrm{L}^{n-1}\,\mathrm{s}^{-1}$ to deduce reaction order from rate constant units.
- **Believing Catalysts Change Equilibrium:** A catalyst speeds up equilibrium attainment; it NEVER shifts equilibrium position or alters K_{eq}.
- **Forgetting Absolute Temperature in Kelvin:** In Arrhenius equations, temperatures must always be converted to Kelvin ($\mathrm{K} = {}^{\circ}\mathrm{C} + 273.15$).

RAPID REVISION MEMORY HOOKS

- **Zero Order:** $t_{1/2} = \frac{[\mathrm{R}]_0}{2k}$ → Linear graph: $[\mathrm{R}]$ vs t (Slope $= -k$).
- **First Order:** $t_{1/2} = \frac{0.693}{k}$ → Linear graph: $\ln[\mathrm{R}]$ vs t (Slope $= -k$).
- **First Order Benchmarks:** $t_{75\%} = 2 \times t_{50\%}$, $t_{87.5\%} = 3 \times t_{50\%}$, $t_{99.9\%} \approx 10 \times t_{50\%}$.
- **Arrhenius Plot:** $\ln k$ vs $1/T$ gives a straight line with $\mathrm{Slope} = -E_a/R$.
- **Effective Collision:** Requires BOTH (1) Energy $\geq E_a$ AND (2) Proper spatial orientation.

9. Concise Final Recap

Chemical Kinetics provides the indispensable quantitative bridge between molecular dynamics and macroscopic chemical change. In summary:

- The **rate of reaction** is defined symmetrically via stoichiometry: $\mathrm{Rate} = -\frac{1}{\nu_\mathrm{R}}\frac{\mathrm{d[R]}}{\mathrm{d}t} = +\frac{1}{\nu_\mathrm{P}}\frac{\mathrm{d[P]}}{\mathrm{d}t}$.
- The **order of reaction** is an empirical quantity (zero, integer, or fractional) representing concentration sensitivity; **molecularity** is a theoretical non-zero integer (1, 2, 3) representing colliding species in an elementary step.
- For complex reactions, the **slowest elementary step** acts as the rate-determining bottleneck.
- **Zero-order** reactions proceed at constant speed until the reactant is exhausted; their half-life is directly proportional to initial concentration.
- **First-order** reactions exhibit exponential decay; their half-life is completely independent of initial reactant concentration. All radioactive decay processes obey first-order kinetics.
- In **pseudo-first-order** reactions, an abundant reagent (such as a solvent) maintains nearly constant concentration, reducing higher-order rate laws to apparent first-order behavior.
- The **Arrhenius equation** ($k = A\,e^{-E_\mathrm{a}/RT}$) mathematically encapsulates the exponential sensitivity of reaction velocity to temperature, explained by the Maxwell-Boltzmann distribution where a $10\,\mathrm{K}$ rise approximately doubles the population of molecules with energy exceeding E_a.
- A **catalyst** accelerates reactions purely kinetically by opening an alternative pathway with a lower E_a; it leaves equilibrium state, ΔG, and ΔH strictly unaltered.
- **Collision Theory** links macroscopic rate constants to microscopic molecular collisions through collision frequency (Z_AB), activation energy (E_a), and steric orientation probability (P).

BOSONANCE